

Silicon Oxidation

References

- S. Wolf, “Silicon processing for the VLSI era”, Lattice Press, Vol. 1, 2nd Edition.
- S. M. Sze, “VLSI Technology”, 2nd Ed. McGraw Hill, 1988

Outline

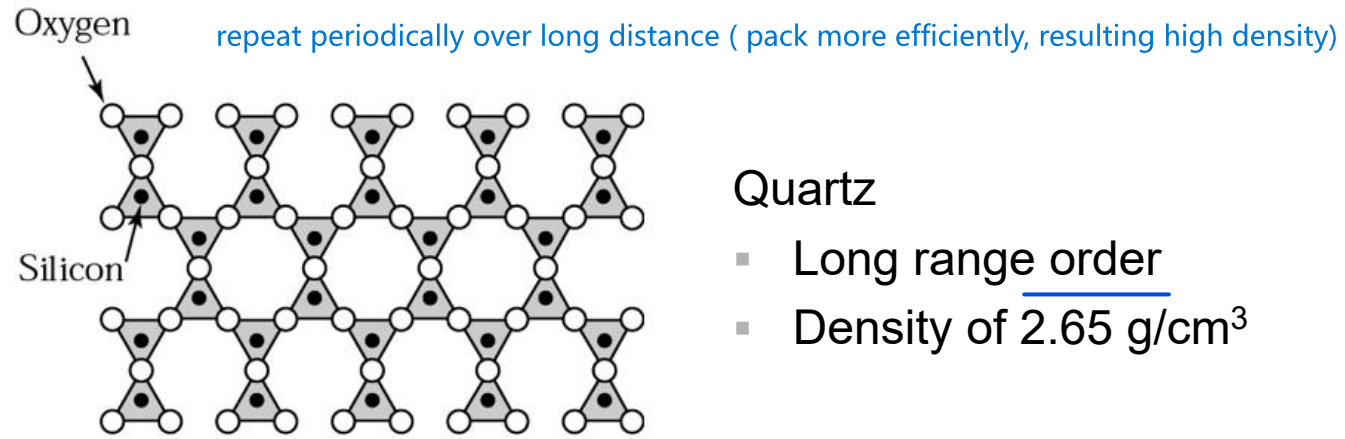
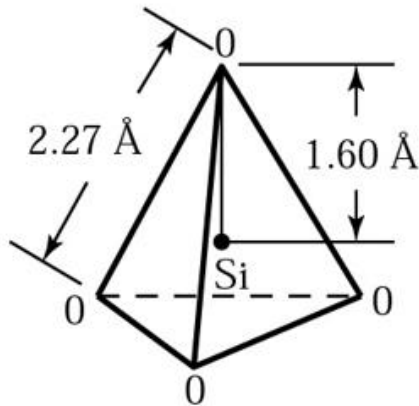
- Properties of Thermal SiO₂ and Applications
- Growth Mechanism & Kinetics: Deal-Grove Model
- Factors Influencing Oxidation
 - Orientation dependence
 - Effects of dopants and impurities
- Thermal SiO₂/Si Interface Properties
 - Dopant re-distribution
 - Oxide charges, Q_{OT} , Q_{IT} , Q_f , Q_M
- Oxidation Technique

Properties of Thermal SiO₂

- Amorphous structure
 - Short range order
 - Density of 2.27 g/cm³
 - Molecular density of 2.3E22 /cm³
- An excellent electrical insulator
 - Energy bandgap of ~8~9 eV
- High breakdown electric field
 - > 10 MV/cm
- Stable, reproducible and high quality SiO₂/Si interface

Property (unit)	Value
Density (g/cm ³)	2.27
Melting point (°C)	~1700
Molecular weight	60.08
Molecules/cm ³	2.3E22
Refractive index	1.46
Dielectric constant	3.9
DC resistivity @25°C (Ohm-cm)	1E16
Energy gap (eV)	~9
Thermal conductivity (W/cm ² °C)	0.014
Etch rate BHF (49%) (nm/min)	100

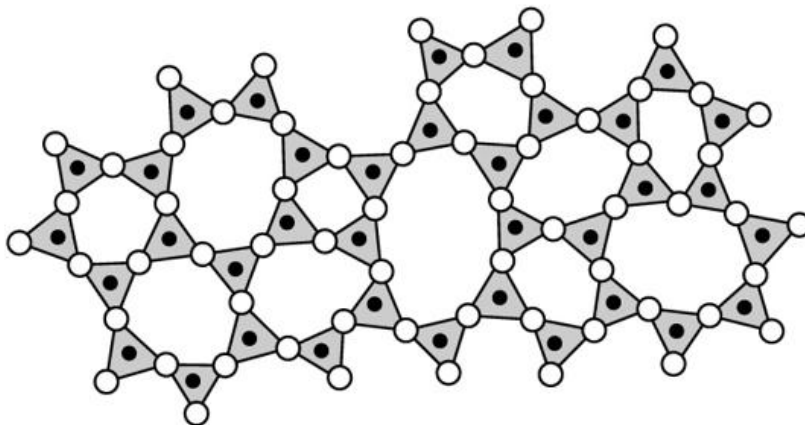
Amorphous structure



Quartz

- Long range order
- Density of 2.65 g/cm³

organized in predictable pattern in short distance, result in lower density

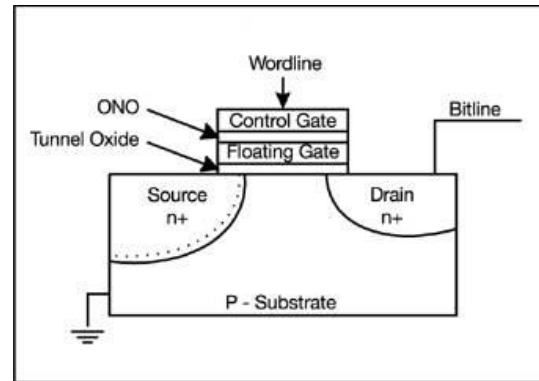
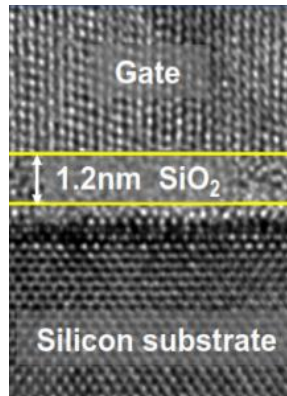


Thermal oxide

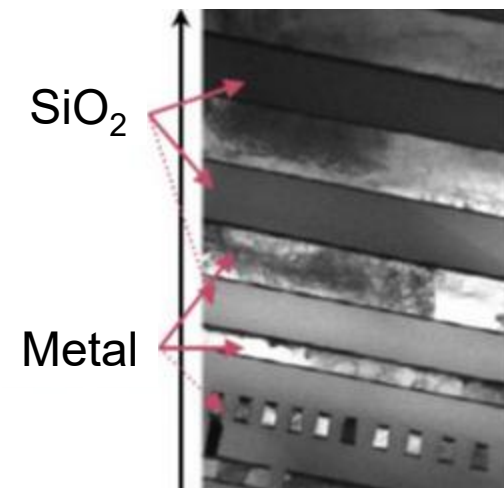
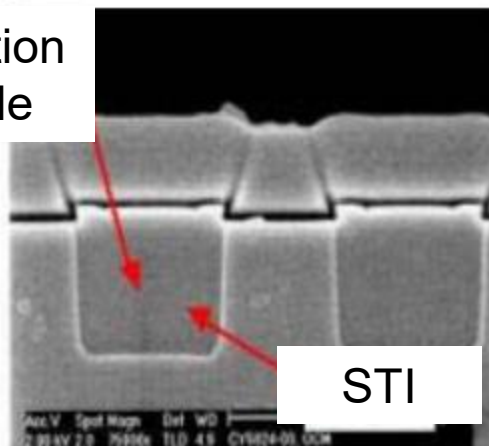
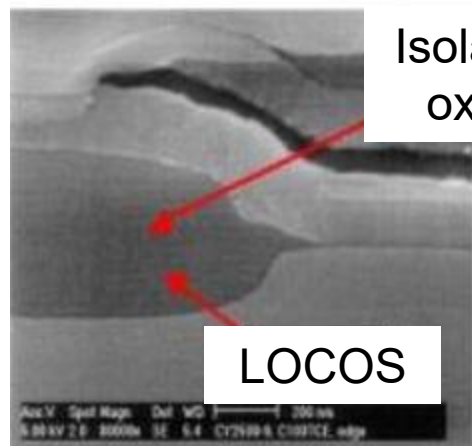
- Short range order
- Density of 2.21 g/cm³

Application: insulator

- Gate oxide in MOSFET, **Tunnel oxide** and ONO in flash memory.

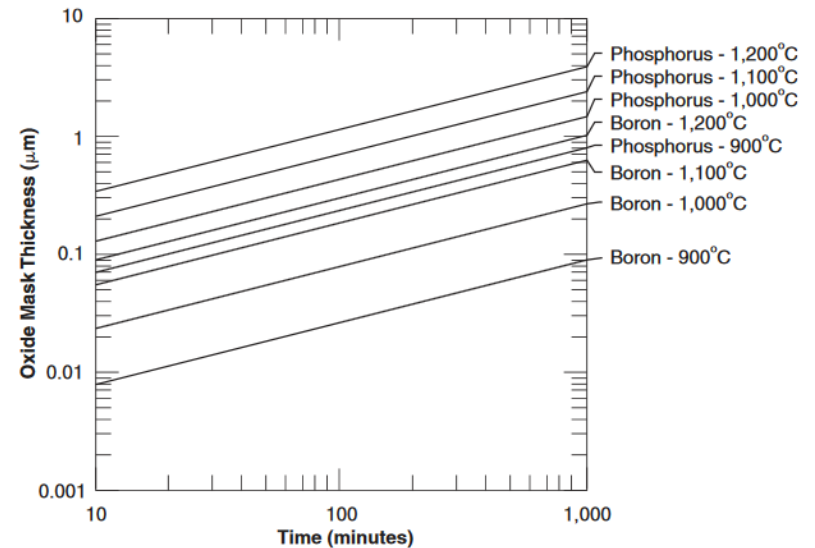
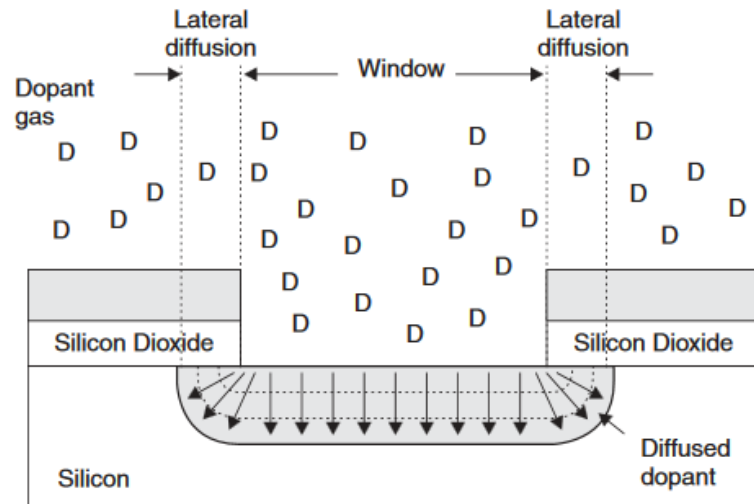


- Isolation oxide in LOCOS and STI in isolating neighboring devices, in interconnect in isolating metal lines.



Application: mask layer

- SiO_2 as selective mask to prevent the dopant diffusion at high temperatures.



- SiO_2 as selective mask against ion implantation.

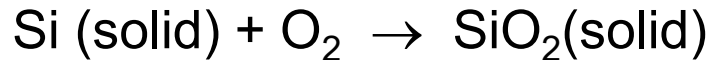
act as over to protect specific area / window concept

Outline

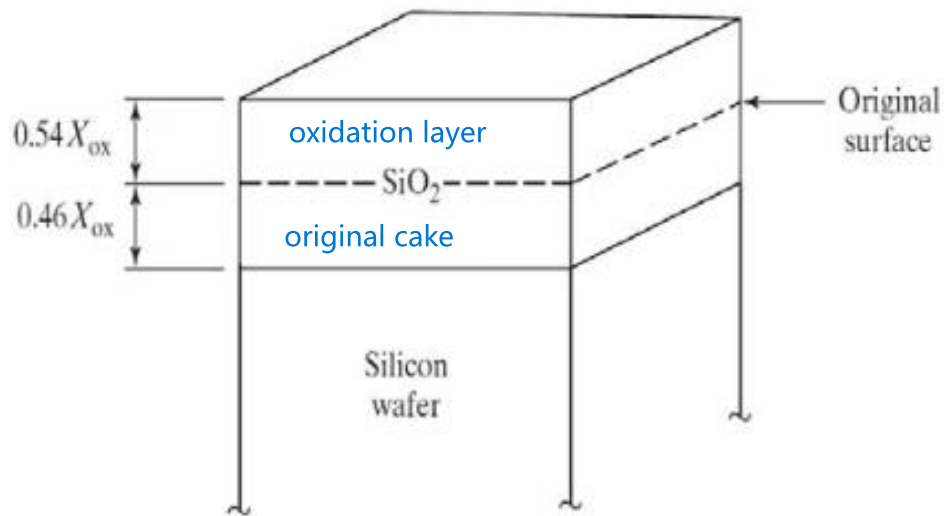
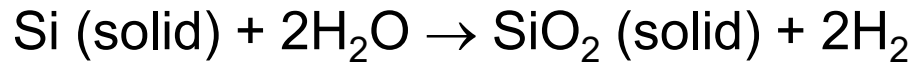
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Wet and Dry Oxidation

- Dry oxidation:



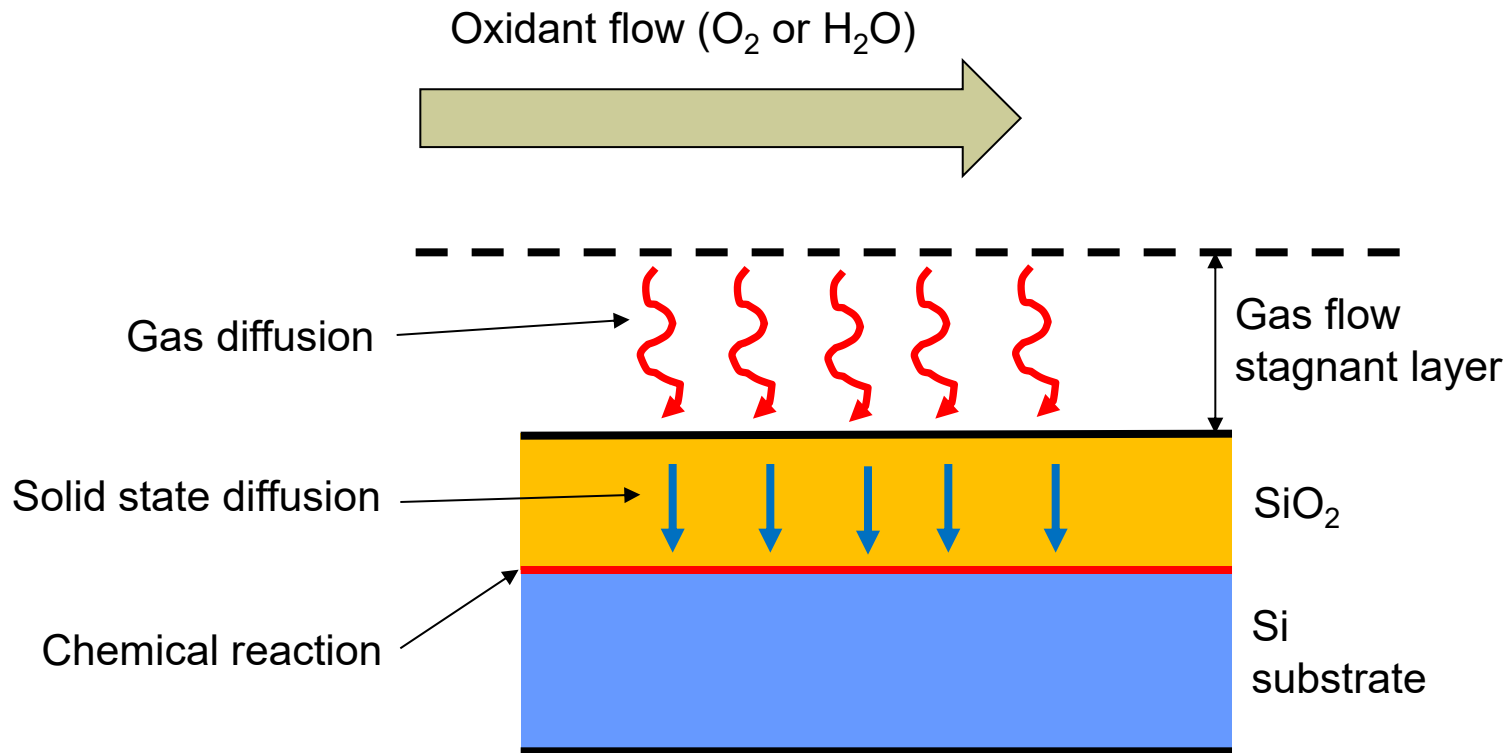
- Wet oxidation:



Volume expands as O₂ is incorporated

Kinetics of Si Oxidation

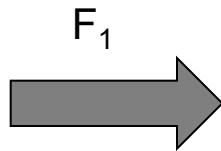
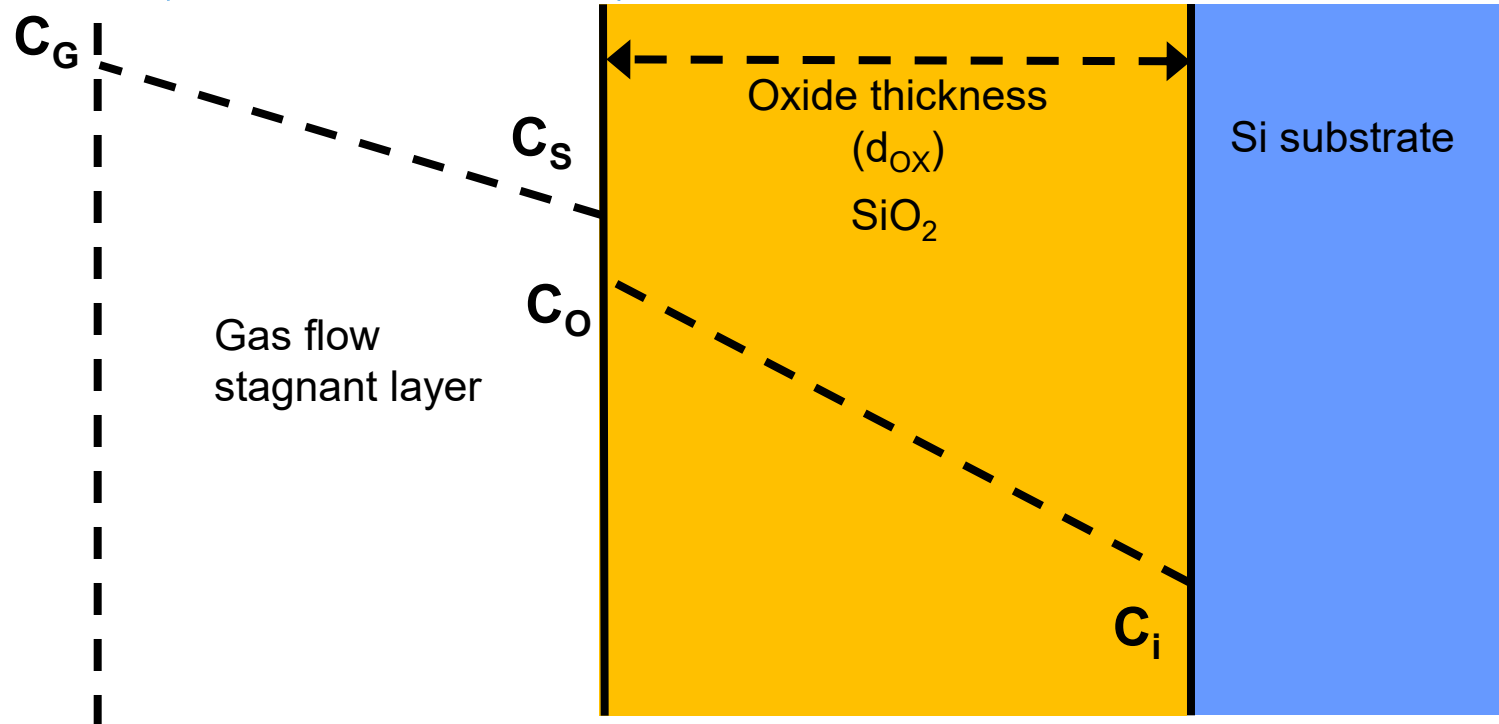
- Oxidation proceeds by the diffusion of oxidizing species through the oxide to the Si-SiO₂ interface, where oxidation reaction occurs.



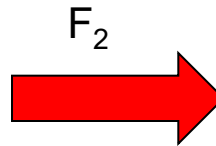
Deal & Grove Model

describe the growth mechanism and kinetic of thermal silicon oxidation
- use to explain how the thickness of SiO_2 layer increase over time

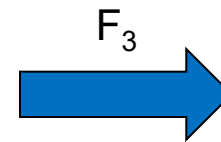
-predict thickness and time



Gas transport
flux



Diffusion flux
through SiO_2



Reaction flux at
interface

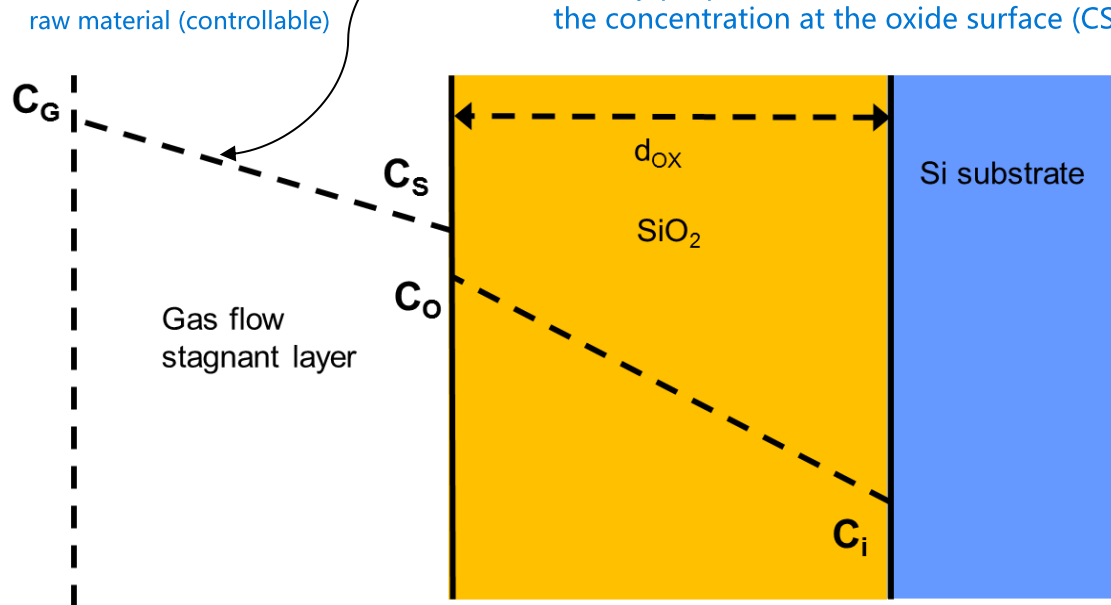
F_1 : from the bulk of the gas phase to the gas/oxide interface

- Assuming the flux of oxidant is proportional to the difference between C_G and C_S , F_1 is given by

$$F_1 = h_G (C_G - C_S)$$

Flux : # molecules / Cm2 sec

directly proportional to the difference between the concentration in the bulk gas (C_G) and the concentration at the oxide surface (C_S), multiplied by the transfer coefficient (h_G)



C_G : oxidant concentration in the bulk of the gas phase

C_S : oxidant concentration at oxide surface of the gas phase

h_G : the gas-phase mass transfer coefficient (cm/sec)

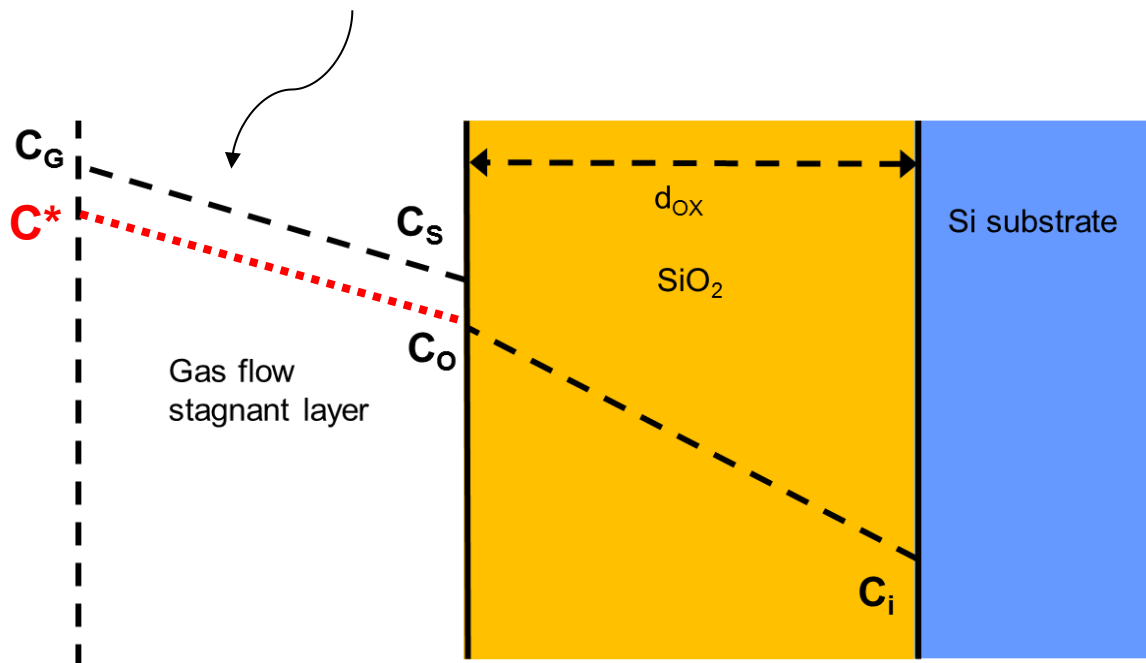
it means how easy the oxidant molecules move from furnace gas to surface wafer

- In the gas phase, we apply the ideal gas law.
*pressure from substances
- To relate the partial pressure (**in the gas phase**) to equilibrium oxidant concentration **in the oxide**, Henry's law is involved.
describe solubility of the gas in solid oxide
- Thus, F_1 can be rewritten as:

$$\Rightarrow C_G = \frac{p_G}{kT} \quad C_S = \frac{p_S}{kT}$$

$$\Rightarrow C^* = H \cdot p_G \quad C_O = H \cdot p_S$$

$$F_1 = h(C^* - C_O) \quad \text{where, } h = \frac{h_G}{HkT}$$



p_G : partial pressure of oxidant in the bulk of the gas phase

p_S : partial pressure of oxidant at the oxide surface of the gas phase

C^* : equivalent oxidant concentration of C_G in the oxide ($=HkTC_G$)

C_O : concentration of oxidant at the oxide surface in the oxide ($=HkTC_S$)

H : Henry's law constant.

h_G : the gas-phase mass-transfer coefficient

C^* = equilibrium limit of how many oxidant the SiO_2 can hold

F_1 is faster than other steps

F_2 : diffusion across the existing oxide toward the silicon

process of oxidant soaking through the wafer to react with

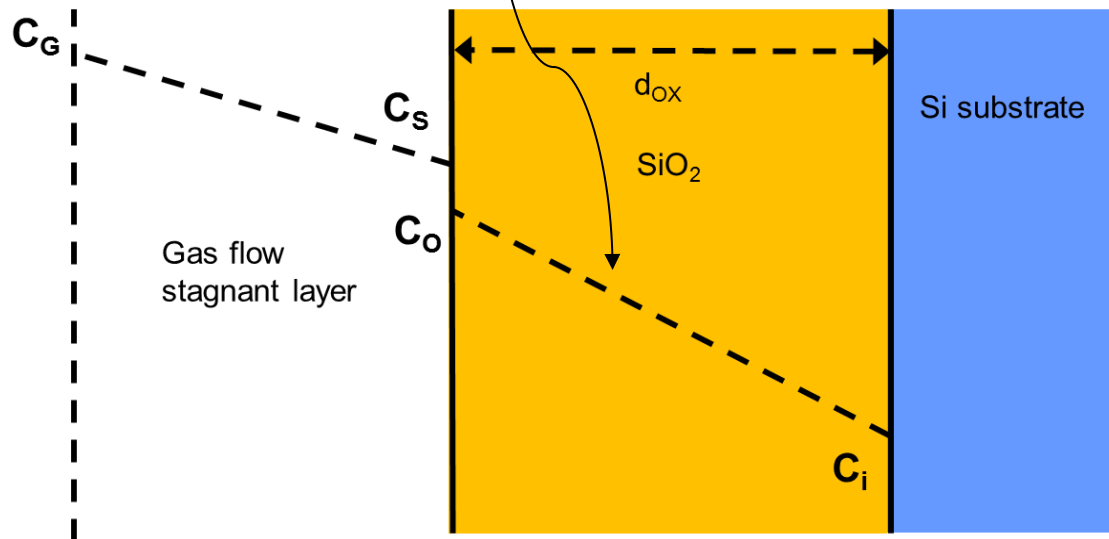
- Assuming F_2 is due to the diffusion of oxidant across the oxide :

$$F_2 = -D \frac{dC}{dx} = D \frac{(C_o - C_i)}{d_{ox}}$$

D : the diffusion coefficient
(cm²/sec)

C_o : concentration of oxidant at the oxide

C_i : concentration of oxidant at the Si/oxide interface
cake



F_3 : reaction with silicon at oxide/silicon interface

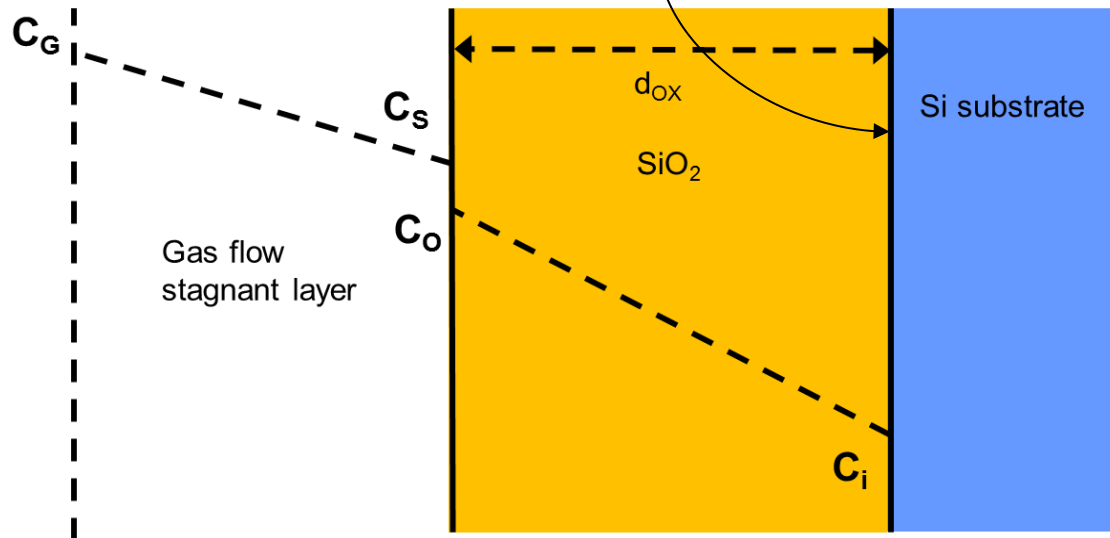
reaction step where the silicon oxid

- Flux corresponding to the Si/SiO₂ interface reaction is assumed to be:

$$F_3 = k_s C_i$$

k = cm/sec reaction rate

C_i : concentration of oxidant at the oxide Si/oxide interface
 k_s : the rate constant of chemical reaction for silicon oxidation at the interface (cm/sec)

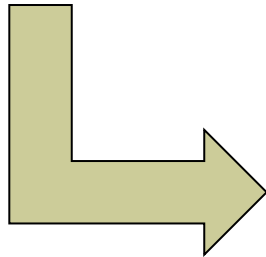
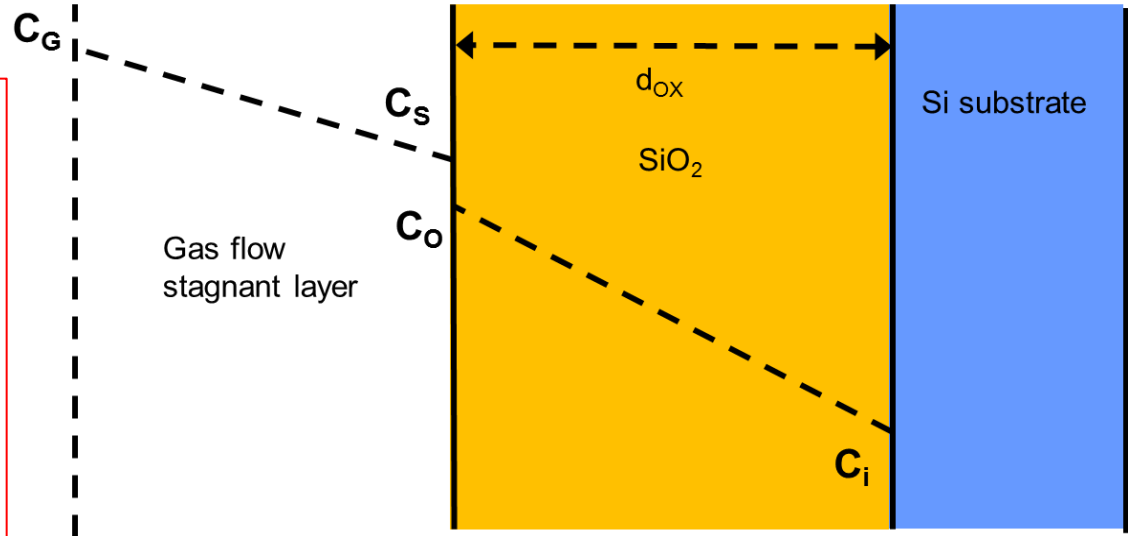


At steady state: $F_1 = F_2 = F_3$

$$F_1 = h(C^* - C_o)$$

$$F_2 = -D \frac{dC}{dx} = D \frac{(C_o - C_i)}{d_{ox}}$$

$$F_3 = k_s C_i$$



Solve C_i and C_o as a function of C^*

Note: C^* or C_G is a controlled process variable (proportional to the input oxidant gas pressure)

$$C_i = \frac{C^*}{1 + \frac{k_s}{h} + \frac{k_s d_{ox}}{D}}$$

$$C_o = \frac{[1 + \frac{k_s d_{ox}}{D}] \cdot C^*}{1 + \frac{k_s}{h} + \frac{k_s d_{ox}}{D}}$$

Two special cases

- Case 1: when D is small or d_{ox} is large (thick)
 - $C_i \rightarrow 0$, $C_O \rightarrow C^*$
 - Diffusion-limited**
 - Oxidant supply to the Si/SiO₂ interface is not enough

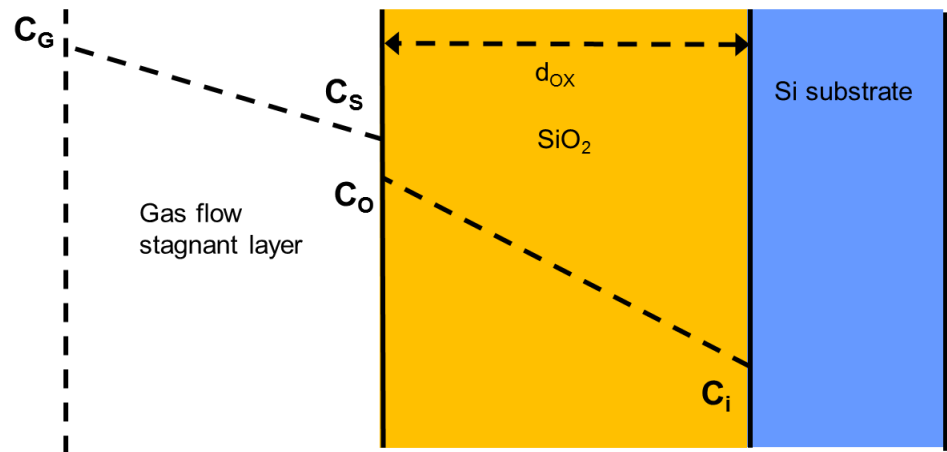
- Case 2: when D is large or d_{ox} is small (thin)

- $$C_i = C_O \rightarrow \frac{C^*}{1 + \frac{k_s}{h}}$$

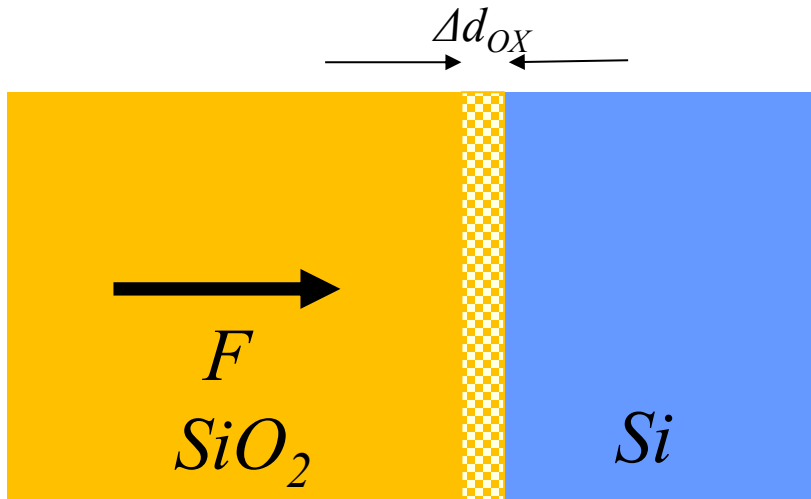
- Reaction-limited**
- An abundant supply of oxidant is provided at the Si/SiO₂

$$C_i = \frac{C^*}{1 + \frac{k_s}{h} + \frac{k_s d_{ox}}{D}}$$

$$C_O = \frac{[1 + \frac{k_s d_{ox}}{D}] \cdot C^*}{1 + \frac{k_s}{h} + \frac{k_s d_{ox}}{D}}$$



How to connect the flux to the oxidation rate?



$$\frac{\Delta d_{ox}}{\Delta t} = \frac{F}{N_1}$$

Flux F : the number of oxidant that crosses a plane per unit area and per unit time
[# of molecules / ($\text{cm}^2 \text{ sec}$)]

$$F \Delta t = N_1 \Delta d_{ox}$$

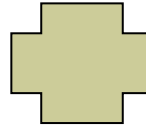
N_1 : Oxidant molecules/unit volume required to form a unit volume of SiO_2
[# of molecules / cm^{-3}]

N_1 : in dry and wet oxidation

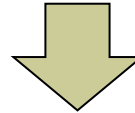
- N_1 is the number of molecules of the oxidizing species in a unit volume of the oxide layer.
- Dry oxidation:
 - $\text{Si (solid)} + \text{O}_2 \rightarrow \text{SiO}_2(\text{solid})$
 - one oxygen molecule (O_2) is added to each silicon dioxide molecule
 - $N_1 = 2.3 \times 10^{22} \text{ cm}^{-3}$ (molecular density of SiO_2)
- Wet oxidation:
 - $\text{Si (solid)} + 2\text{H}_2\text{O} \rightarrow \text{SiO}_2 (\text{solid}) + 2\text{H}_2$
 - two water molecules (H_2O) are added to each silicon dioxide molecule
 - $N_1 = 4.6 \times 10^{22} \text{ cm}^{-3}$

Oxidation rate: **linear-parabolic relationship**

$$N_1 \frac{dd_{ox}}{dt} = F_3 = k_s C_i = \frac{k_s C^*}{1 + \frac{k_s}{h} + \frac{k_s d_{ox}}{D}}$$



Initial condition : **when** $t = 0$, $d_{ox} = d_i$



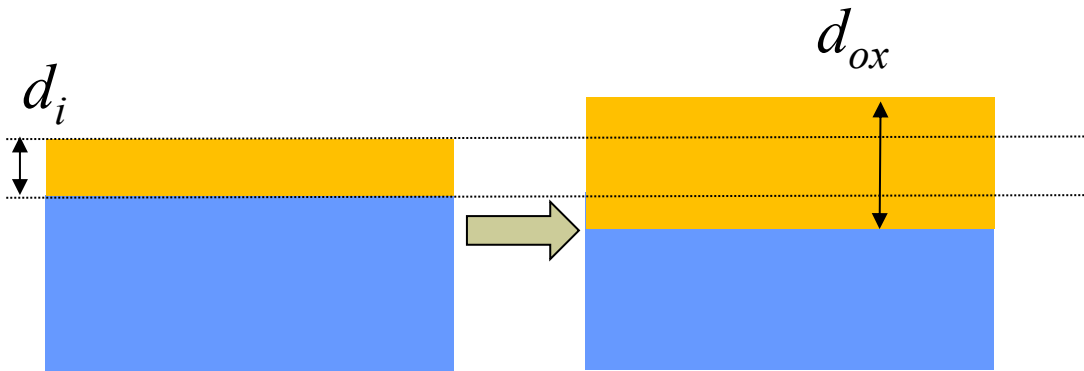
$$d_{ox}^2 + A \cdot d_{ox} = B(t + \tau)$$

where,

$$A = 2D\left(\frac{1}{k_s} + \frac{1}{h}\right)$$

$$B = \frac{2DC^*}{N_1}$$

$$\tau = \frac{d_i^2 + Ad_i}{B}$$



Some notes:

- $h \gg k_s$ for a typical oxidation condition.
- τ represents a time shift to account for the presence of initial oxide layer d_i .

Oxidation rate: two special cases

$$d_{ox} / \left(\frac{A}{2} \right) = \left(1 + \frac{t + \tau}{A^2 / 4B} \right)^{\frac{1}{2}} - 1$$

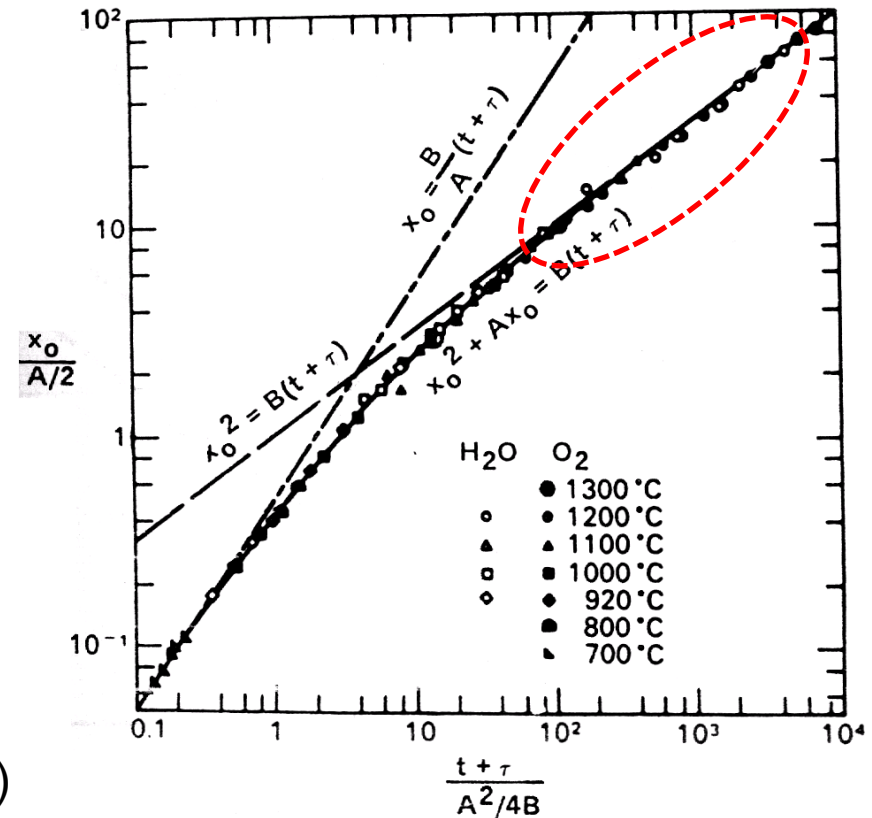
- Case 1: long time oxidation ($t \gg \tau$),

$$d_{ox}^2 \cong B \cdot t$$

$$B = \frac{2DC^*}{N_1}$$

$$d_{ox} \cong \sqrt{Bt}$$

- B: parabolic rate constant
- B mainly relates to D (diffusion coefficient of oxidant through SiO_2)
- Long oxidation time $\rightarrow$ large d_{ox} value (thick oxide)
- Oxidation is limited by diffusion of oxidant through SiO_2 .



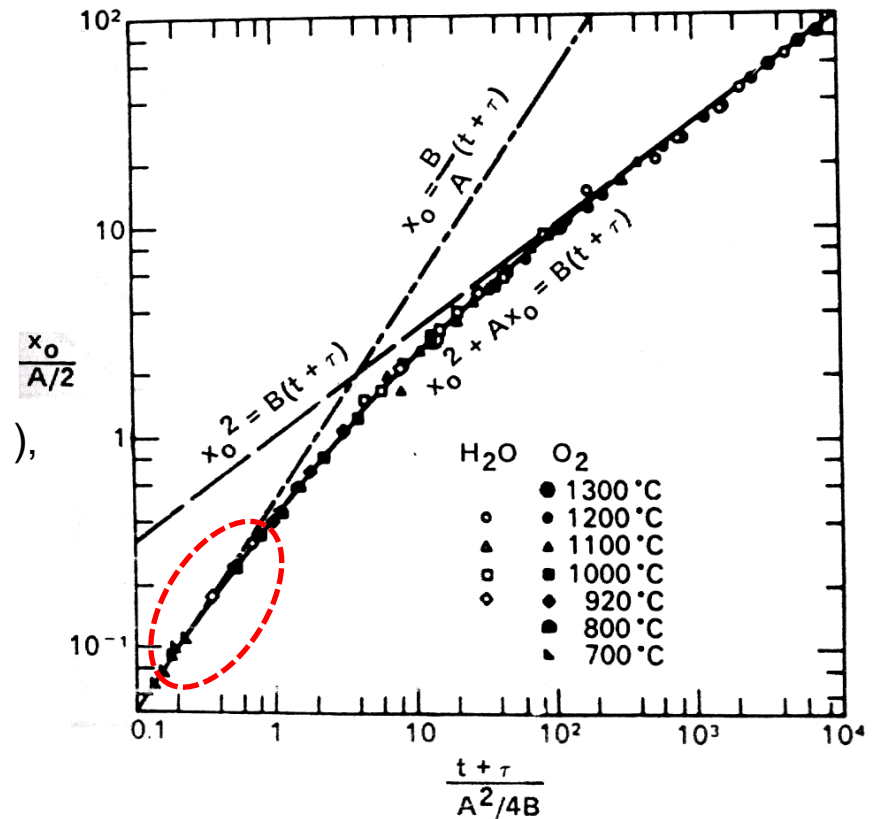
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$$d_{ox} / \left(\frac{A}{2} \right) = \left(1 + \frac{t + \tau}{A^2 / 4B} \right)^{\frac{1}{2}} - 1$$

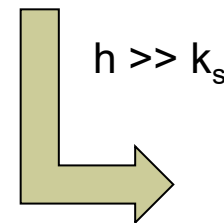
- Case 2: Short time oxidation ($t + \tau \ll A^2 / 4B$),

$$d_{ox} = \frac{B}{A} (t + \tau)$$

- $\frac{B}{A}$: linear rate constant
- (B/A) relates to k_s (rate of chemical reaction at Si/SiO₂ interface)
- Short oxidation time $\rightarrow$ small d_{ox} value (thin oxide)
- Oxidation is limited by chemical reaction at Si/SiO₂.



$$\frac{B}{A} \left(= \frac{k_s h}{k_s + h} \cdot \frac{C^*}{N_1} \right)$$



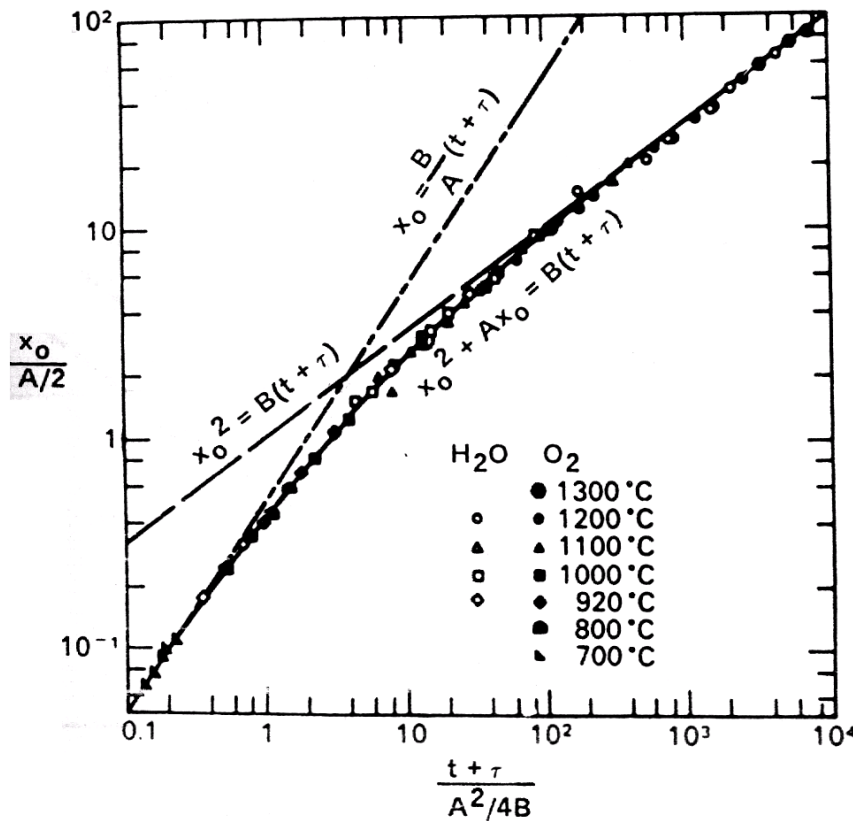
$$\frac{B}{A} = k_s \left(\frac{C^*}{N_1} \right)$$

Oxidation rate: rate decrease with oxide thickness

$$d_{ox}^2 + A \cdot d_{ox} = B(t + \tau)$$

$$2d_{ox}d(d_{ox}) + Ad(d_{ox}) = B(dt)$$

$$\frac{d(d_{ox})}{dt} = \frac{B}{2d_{ox} + A}$$



D-G model parameters: B and B/A

$$B = \frac{2DC^*}{N_1}$$

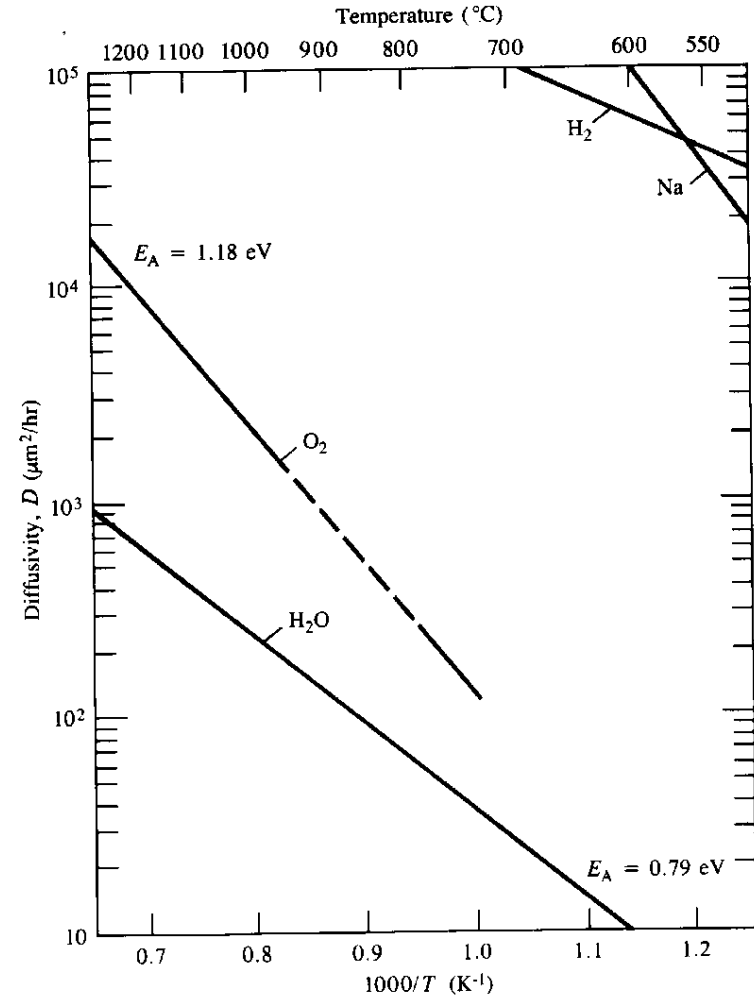
$$D \propto \exp\left(-\frac{E_A}{kT}\right)$$

E_A : activation energy for oxidant diffusion through SiO_2 .

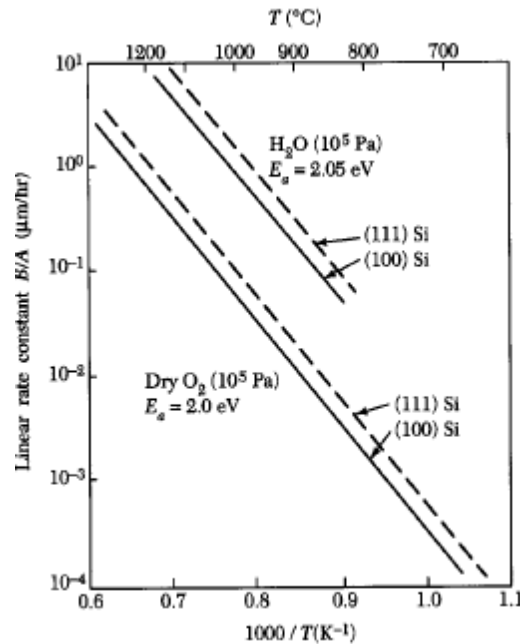
$$\frac{B}{A} = k_s \left(\frac{C^*}{N_1}\right)$$

$$k_s \propto \exp\left(-\frac{E'_A}{kT}\right)$$

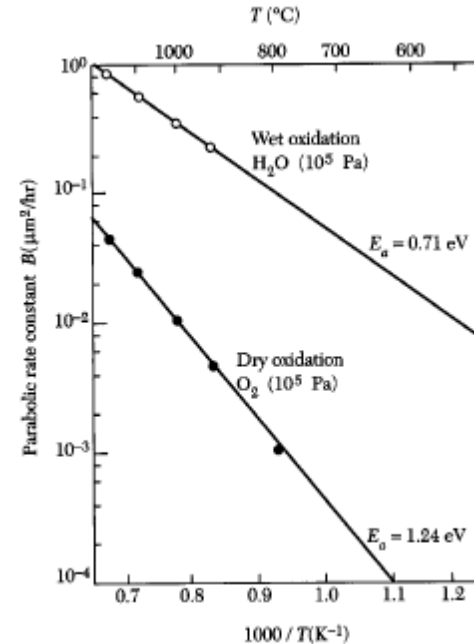
E'_A : activation energy for oxidation reaction at Si/SiO_2 interface.



Linear rate constant B/A



Parabolic rate constant B



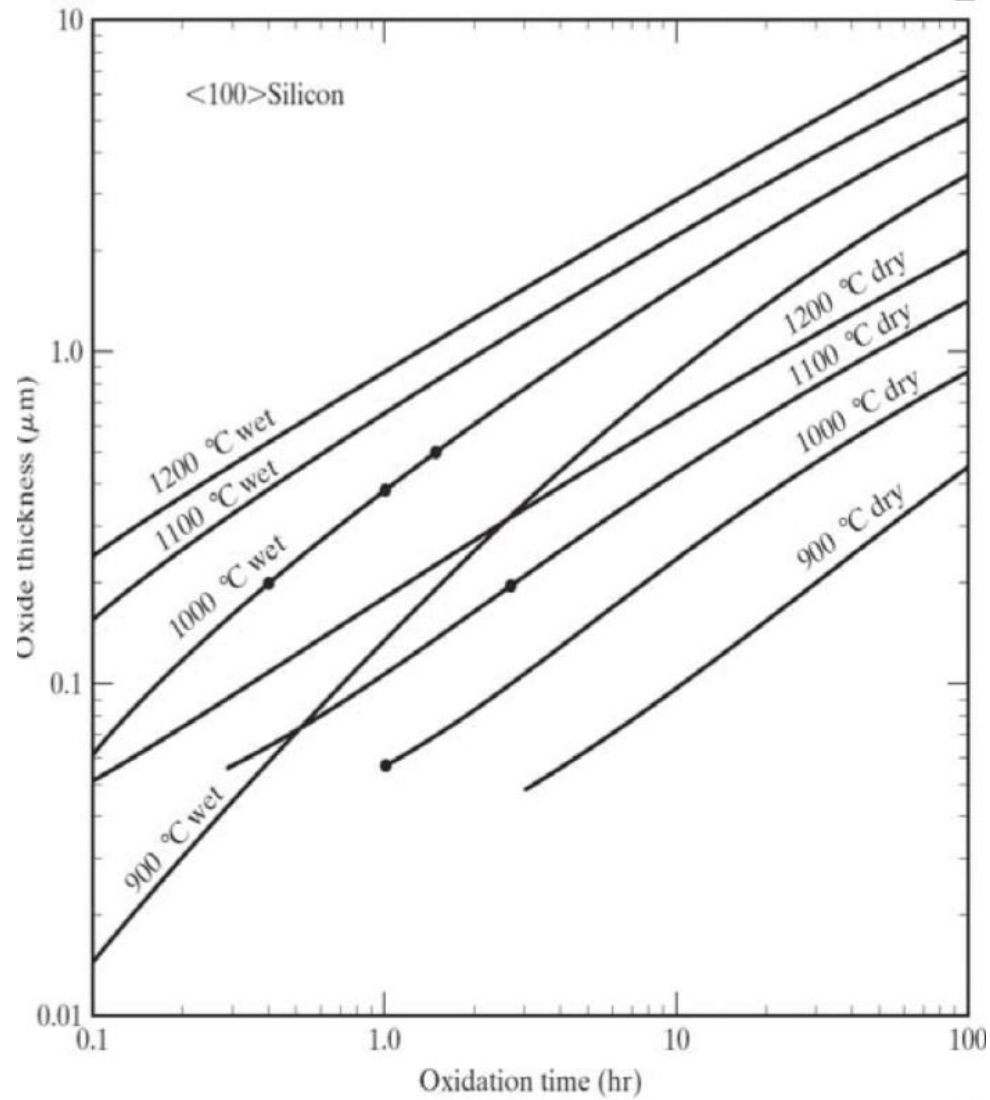
Note:

- Though D of water $<$ D of oxygen in oxide, B for wet oxidation is substantially larger than for dry oxidation. (C^* for O_2 : $5.2 \times 10^{16} \text{ cm}^{-3}$ at 1000°C and C^* for H_2O : $3.0 \times 10^{19} \text{ cm}^{-3}$ at 1000°C)

	B/A ($\mu\text{m/hr}$)	B ($\mu\text{m}^2/\text{hr}$)	τ (hr)
1000°C Dry	0.071	0.0117	0.37
1000°C Wet	1.27	0.287	0
1200°C Wet	14.4	0.72	0

- D-G model works well for wet oxidation, a zero " τ " for $d_i=0$.
- For dry oxidation, a non-zero " τ " is required for $d_i=0$ implies the D-G model underestimates the thickness of oxide in dry oxidation.

Oxidation charts



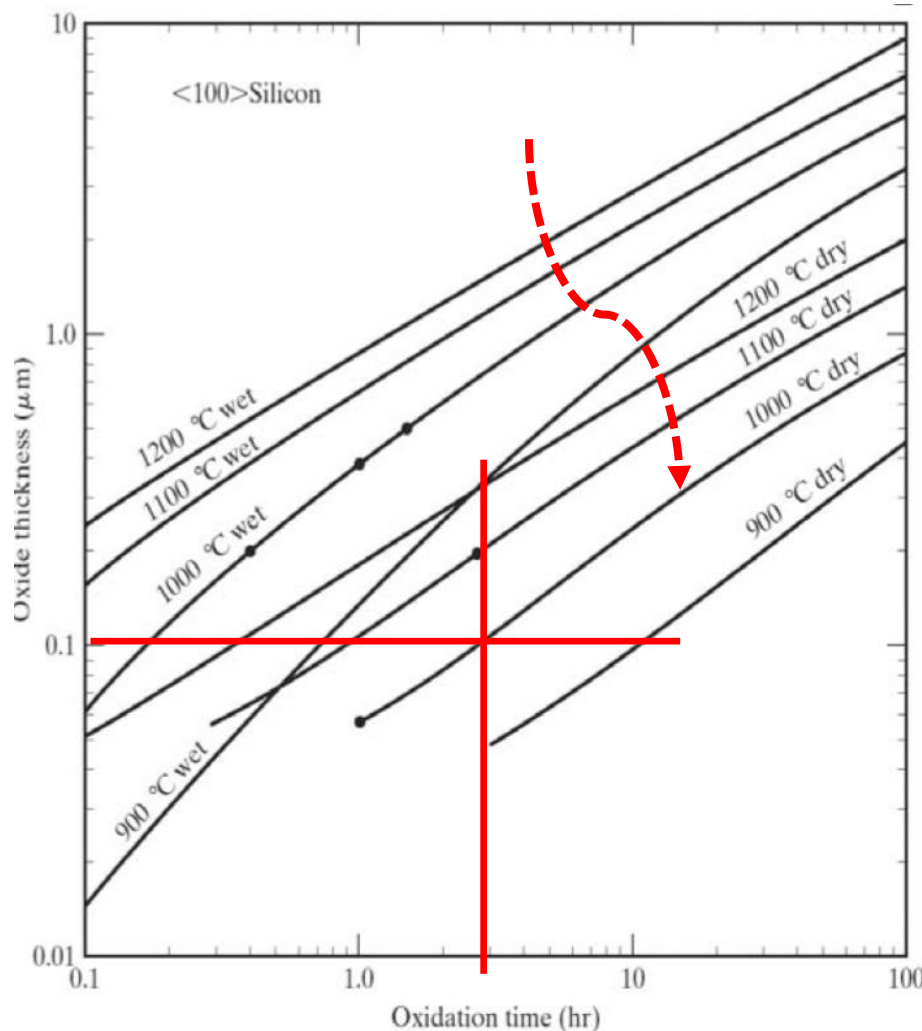
Example: how to estimate oxidation time or oxide thickness

A $\langle 100 \rangle$ silicon wafer has a 1000 Å oxide on its surface.

1. How long did it take to grow this oxide at 1000 °C in dry oxygen?
2. The wafer is put back in the furnace in wet oxidation ambient at 1000 °C. How long will it take to grow an addition 3000 Å of oxide?

Method 1: using oxidation chart

A $\langle 100 \rangle$ silicon wafer has a 1000 Å oxide on its surface.



1. How long did it take to grow this oxide at 1000 °C in dry oxygen?

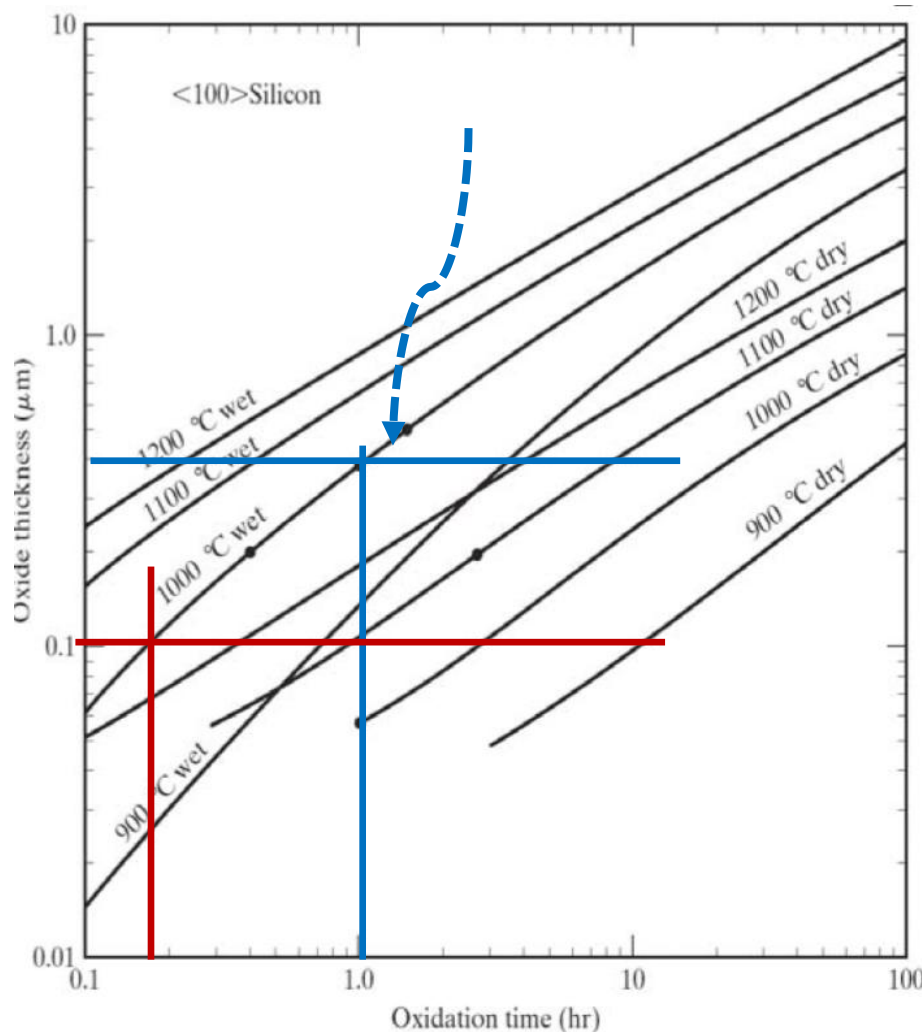
Solution:

- (i) Find the line for 1000 °C dry oxidation
- (ii) Find 1000 Å ($0.1 \mu\text{m}$) in y-axis and draw a line parallel to x-axis
- (iii) Find the intersect point, and draw a vertical line and intersect with x-axis
- (iv) Read t from intersection with x-axis, in this case, ~2.9 hours

Thus, it would take ~**2.9 hours** to grow **$0.1 \mu\text{m}$** oxide in dry oxygen at 1000 °C.

A $\langle 100 \rangle$ silicon wafer has a 1000 Å oxide on its surface.

2. The wafer is put back in the furnace in wet oxidation ambient at 1000 °C. How long will it take to grow an addition 3000 Å of oxide?



Solution:

- Find the curve for 1000C wet oxidation.
- The total oxide thickness at the end of oxidation is **0.4 μm** which would require **1.0 hrs** to grow if there was no oxide on the surface to begin with.
- However the wafer “thinks” it has already been in the furnace for **0.15 hrs** (corresponding to 0.1 μm oxide).
- Thus the additional time needed to grow the **0.3 μm** oxide is **1.0 - 0.15 = 0.85 hrs**.

Method 2: Using D-G model with B and B/A parameters

Try it yourself using the D-G model equation and its parameters.

$$d_{ox}^2 + A \cdot d_{ox} = B(t + \tau)$$

	B/A ($\mu\text{m/hr}$)	B ($\mu\text{m}^2/\text{hr}$)	τ (hr)
1000°C Dry	0.071	0.0117	0.37
1000°C Wet	1.27	0.287	0
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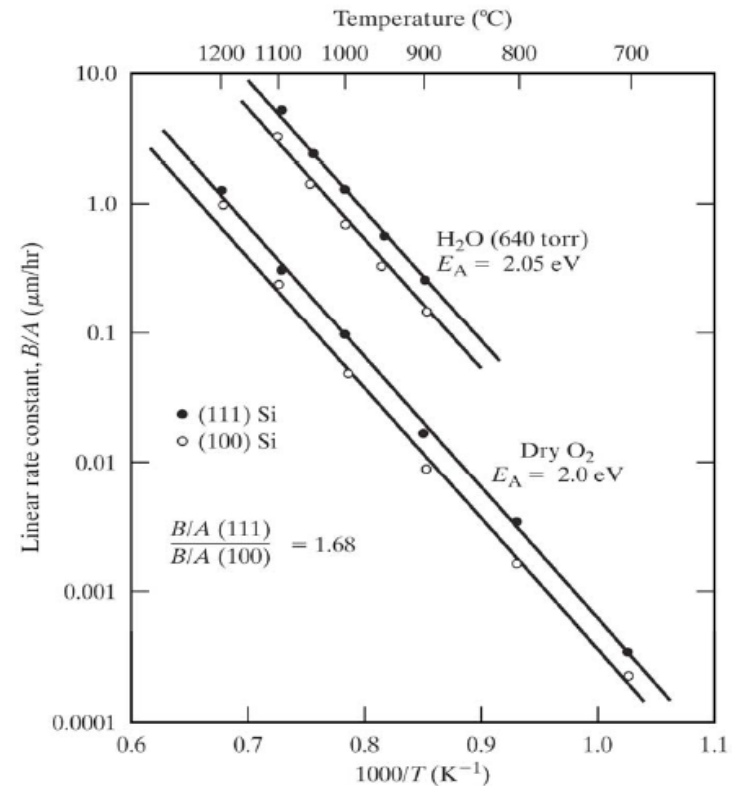
Orientation dependence

- Linear rate constant B/A :
 - Interface reaction limited: k_s
 - Reaction depends on the area density of available silicon atoms
 - Orientation dependence: (111) is faster than (100)

$$\frac{B}{A} = k_s \left(\frac{C^*}{N_1} \right)$$

- Parabolic rate constant B :
 - Diffusion limited: D (oxidant diffusion through SiO_2).
 - Independent with silicon surface orientation.

$$B = \frac{2DC^*}{N_1}$$

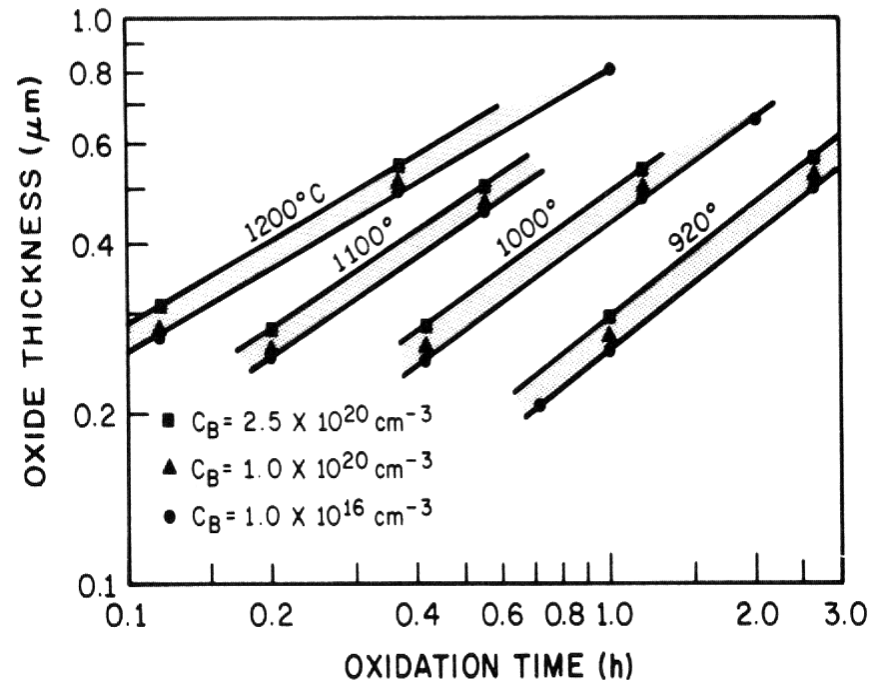


Effects of dopants: boron

- High concentration of boron ($>1\text{E}20\text{ cm}^{-3}$) enhances the oxidation rate.

⇒ is related to boron segregation in SiO_2 .

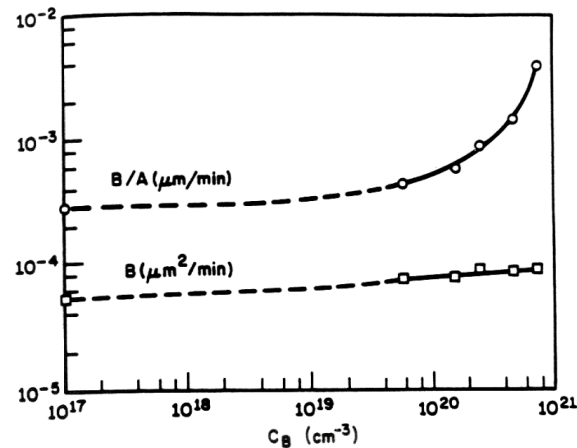
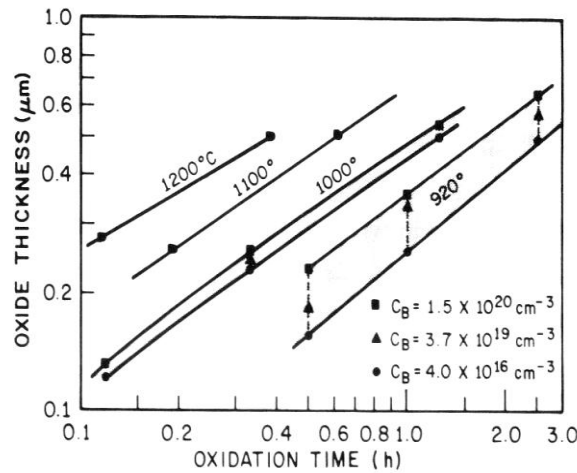
- When the boron segregates into the oxide and remains there ($m < 1$), the bond structure in the silica weakens. This weakened structure permits an increased incorporation and diffusivity of oxidant through the oxide, thus enhancing the oxidation rate.
- The **constant B** (reflecting the diffusion-limited) is increased, while constant B/A is almost not changed.
- Other Group III dopants (Ga) do not cause enhanced oxidation rate. Ga will rapidly diffusive away from the interface.



$$B = \frac{2DC^*}{N_1}$$

Effects of dopants: phosphorous

- Enhanced oxidation rate is only observed at low temperature. Enhancement diminishes at high temperatures.

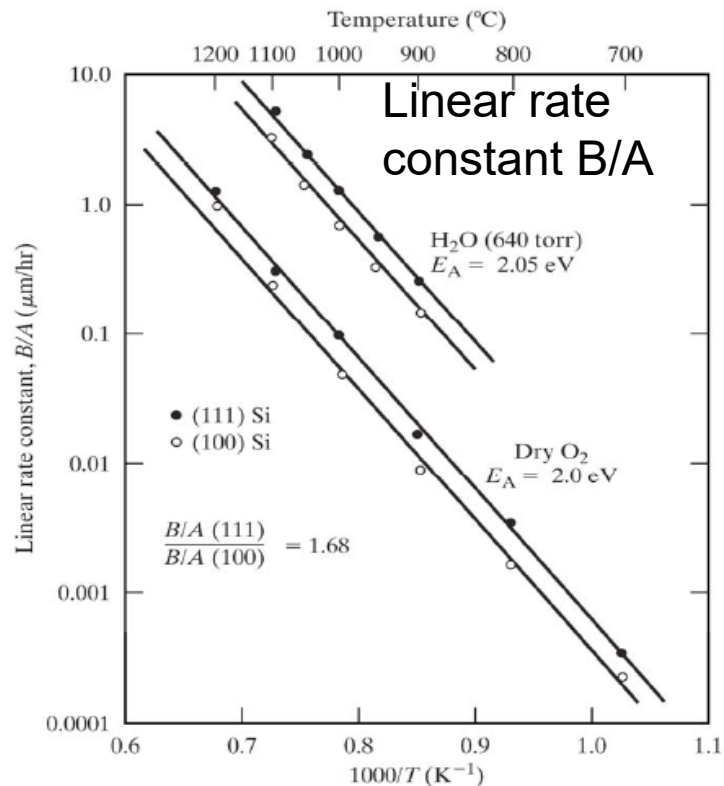
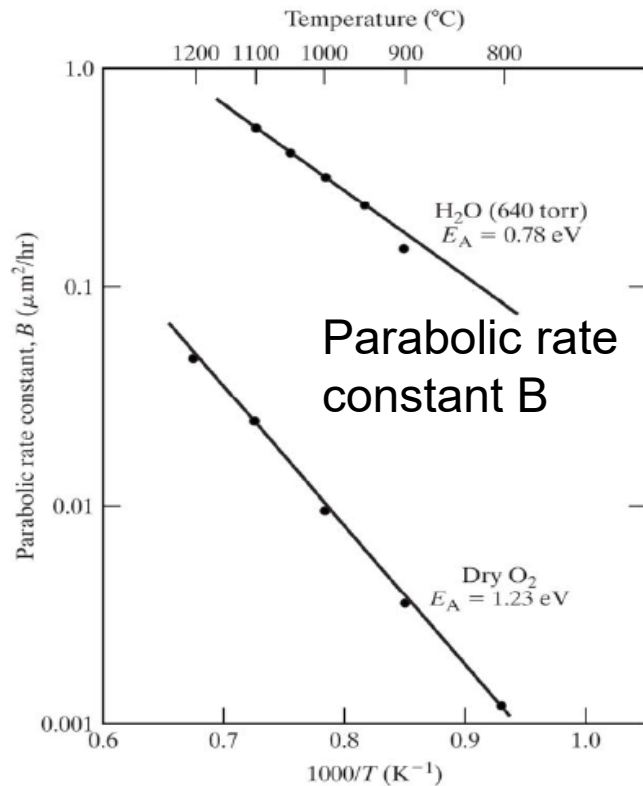


$$\frac{B}{A} = k_s \left(\frac{C^*}{N_1} \right)$$

- Phos will pile up at the Si surface ($m > 1$) and very little is incorporated into the oxide.
- Thus, B/A (relates to the interface chemical reaction) increases substantially at high concentration of P, while parameter B (relates to the oxidant diffusion through oxide) is almost unchanged.
- Arsenic also shows similar phenomena.

Effects of other impurities: water vapor

- Both B and B/A increase with water vapour
- For dry oxidation, H_2O content should be kept below **1 ppm**.



Effects of other impurities: Cl and crystal damage

Chlorine (Cl)

- Cl is intentionally introduced into the oxidation ambient to improve both the oxide and the underlying silicon properties, because Cl has a **gettering effect** by converting certain impurities (such as sodium,) into volatile chlorides or electrically inactive state.
- Cl is typically introduced into dry O₂ ambient in the form of Cl₂ gas, HCl, TCE, or TCA, with a range from 1 to 10 %. (Too high concentrations of Cl at high temperature can pit the silicon surface.)
- Addition of Cl leads to an increase of oxidation rate by 5 to 30 %.

Crystal Damage

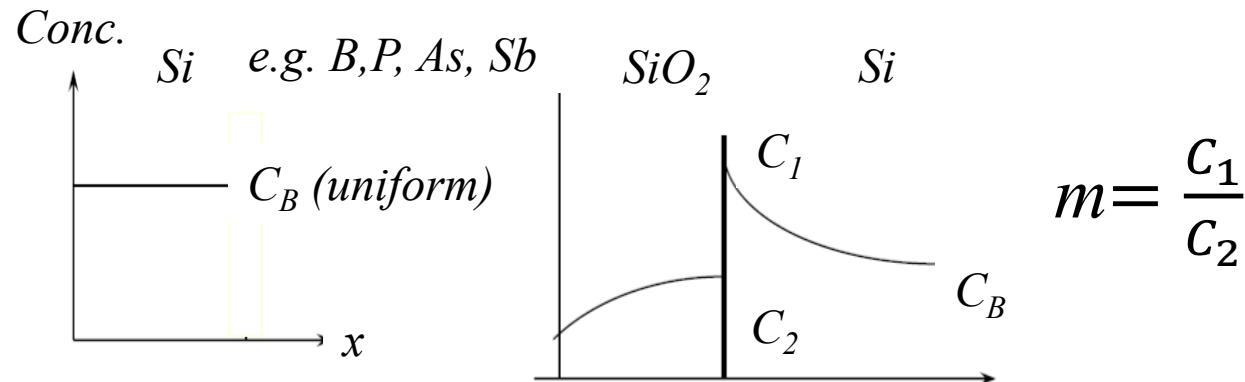
- Crystal damage on the silicon surface is mainly caused by ion implantation or RIE etching at high power level.
- For B, P, As, and Sb implantation, enhancement of oxidation rate was found
 - ⇒ but **two effects of dopant and damage are mixed.**
- For Si and Ge implantation, retardation of oxidation rate was found.
 - ⇒ probably true effect of **crystal damage.**
- RIE at high power also shows **retardation effect.**

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- Oxidation Technique

Dopant Redistribution

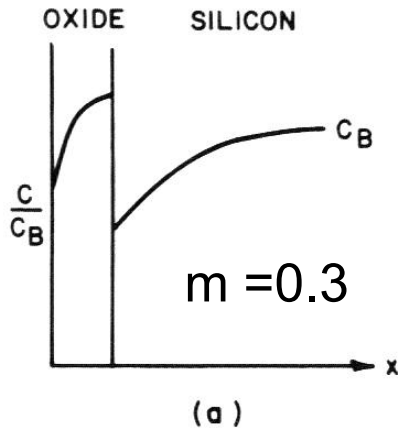
- As oxidation proceeds, the interface advances into the silicon, and dopants will redistribute at the interface until its chemical potential is the same on each side of the interface.
- This redistribution may result in an abrupt change in impurity concentration across the interface.



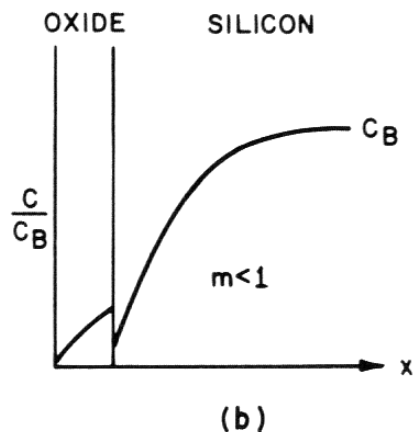
$$m = \frac{\text{equilibrium concentration of the dopant in silicon}}{\text{equilibrium concentration of the dopant in oxide at the interface}}$$

Dopant segregation between Si/SiO₂: 4 cases of interest

$$m < 1$$

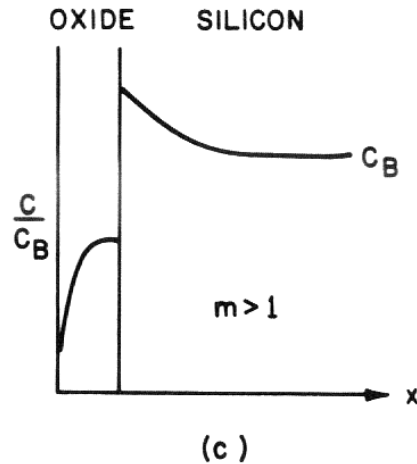


- Segregation into the oxide with **slow diffusion**
- Dopant depleted at the Si surface
- Dopant: B ($0.1 < m < 1$)

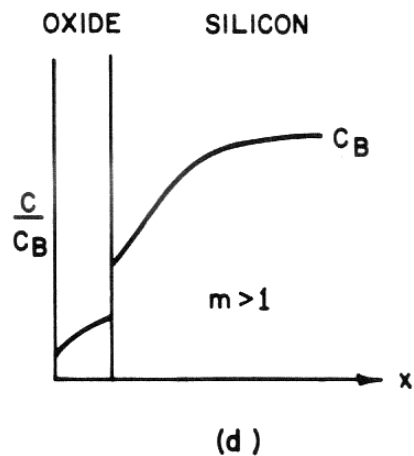


- Segregation into the oxide with **fast diffusion**
- Dopant depleted at Si surface
- Dopant: boron in H₂ ambient

$$m > 1$$

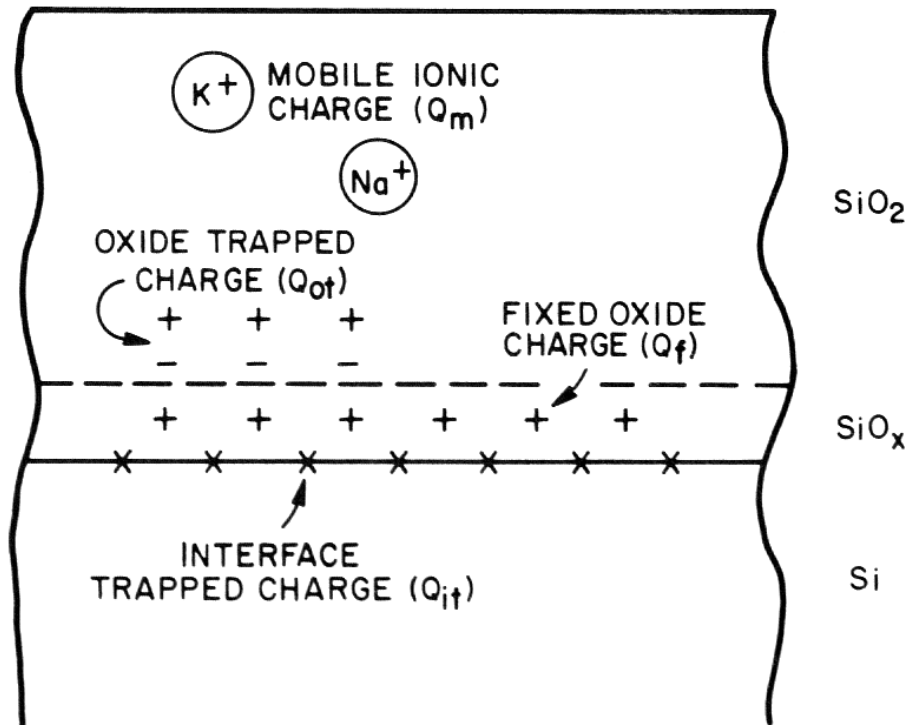


- Segregation into the silicon with **slow diffusion**
- Dopant pile up at Si surface
- Dopant: P, As, Sb ($m > 10$)



- Segregation into the silicon with **fast diffusion**
- Dopant is depleted at surface
- Dopant: Ga

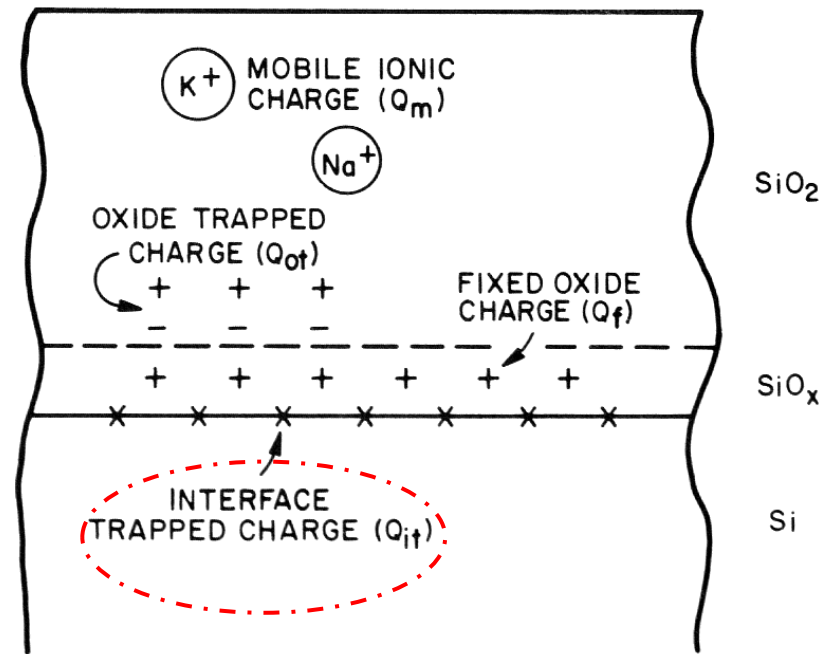
Oxide Charges



Interface-trapped charge Q_{it}

- Originates from structural defects related to oxidation process, metallic impurities, or bond breaking processes.
- Locates at the Si/SiO₂ interface with energy states in the silicon forbidden bandgap.
- A low temperature hydrogen anneal (post-metallization anneal) is effective in reducing Q_{it} .

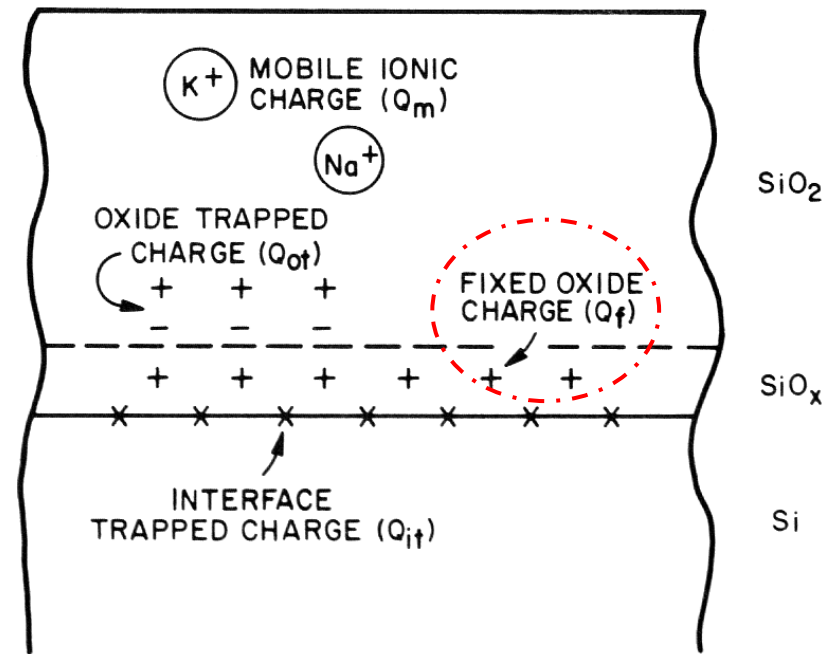
⇒ usually annealing in forming gas (H₂-N₂) ambient in temperature range of 350 ~ 500 °C



D_{it} of $10^{10}/\text{cm}^2.\text{eV}$

Fixed oxide charge Q_f

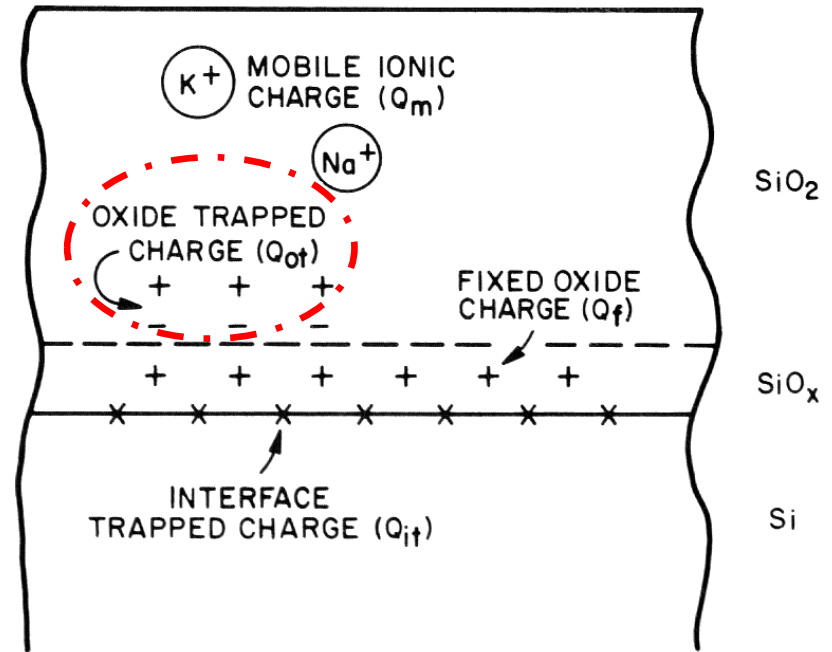
- Originates from **nonstoichiometric region** (strained region) of SiO_2 near the Si/SiO_2 interface (SiO_x where $1 < x < 2$)
- Immobile under an applied electric field. (cannot be charged or discharged)
- For electrical measurement, Q_f is considered as a charge sheet at the Si/SiO_2 interface. (typically measured by ΔV_{FB} in C-V curve.)



- Q_f in a range from $10^{10}/\text{cm}^2 \sim 10^{12}/\text{cm}^2$ depending on oxidation and annealing conditions as well as silicon wafer orientation.
- Orientation dependence : $Q_f(100) < Q_f(111)$, relating to the number of available bonds per unit area of silicon surface.
- The last high temperature thermal treatment determines Q_f .
(Usually, a higher temperature produces a lower value of Q_f)
 - Fast cooling from high temperature oxidation to prevent oxidation at low temperature.
 - When low temperature oxidation is inevitable (e.g. very thin gate oxide formation), in-situ subsequent high temperature annealing in an inert gas ambient needs to be added.

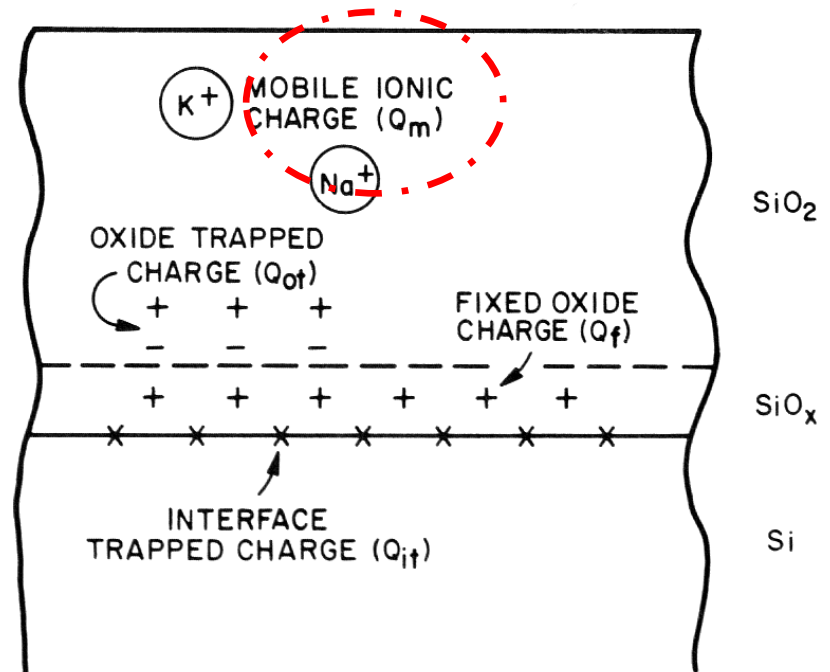
Oxide trapped charge Q_{ot}

- Positively (hole) or negatively (electron) trapped charge in the bulk oxide
- Associated with defects in the SiO_2 , and may result from X-ray radiation or hot electron injection.
- Can be annealed out (not perfectly).
- In the range from $10^{10} \sim 10^{12} / \text{cm}^2$.
- Trapping and de-trapping cause hysteresis in C-V curve.



Mobile ionic charge Q_M

- Attributed to alkali ions such as sodium, potassium, and heavy metals
- In the range from $10^{10} \sim 10^{12} / \text{cm}^2$
- Related to the processing environment, i.e. cleanliness of the oxidation process including such components of the process as :
 - the furnace tube,
 - processing chemicals,
 - oxidizing ambient,
 - wafer handling, and so on.
- Addition of chlorine during oxidation neutralizes Na ions.
- Can be measured by a temperature-bias stress test.



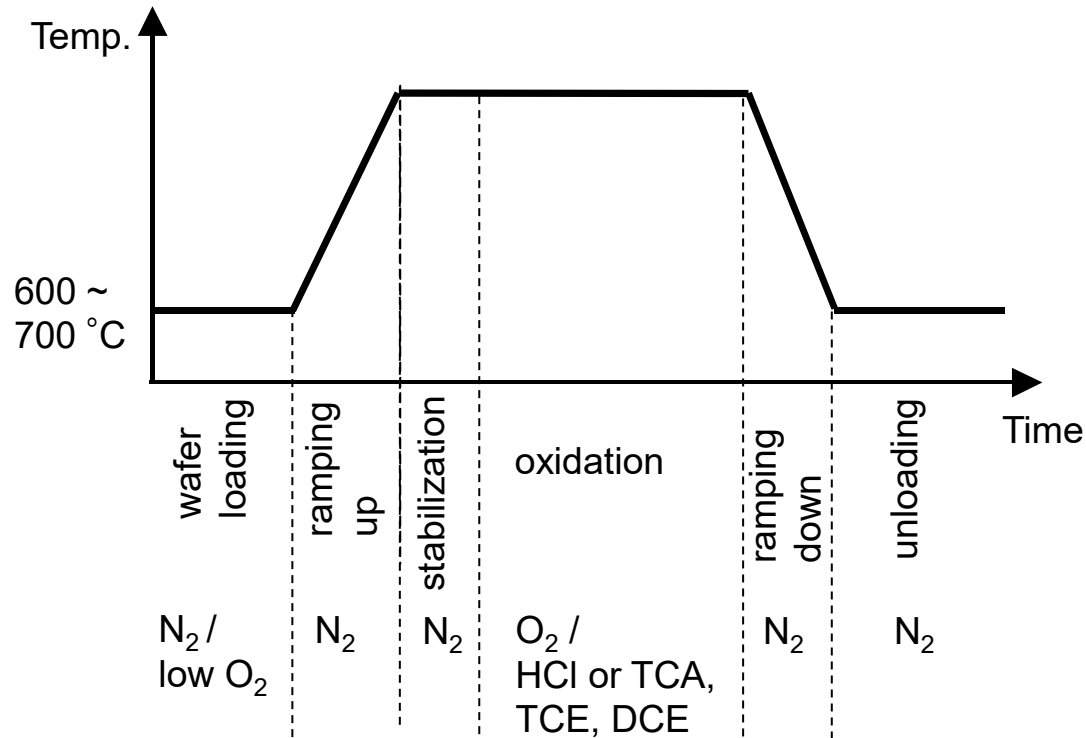
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Oxidation Recipe

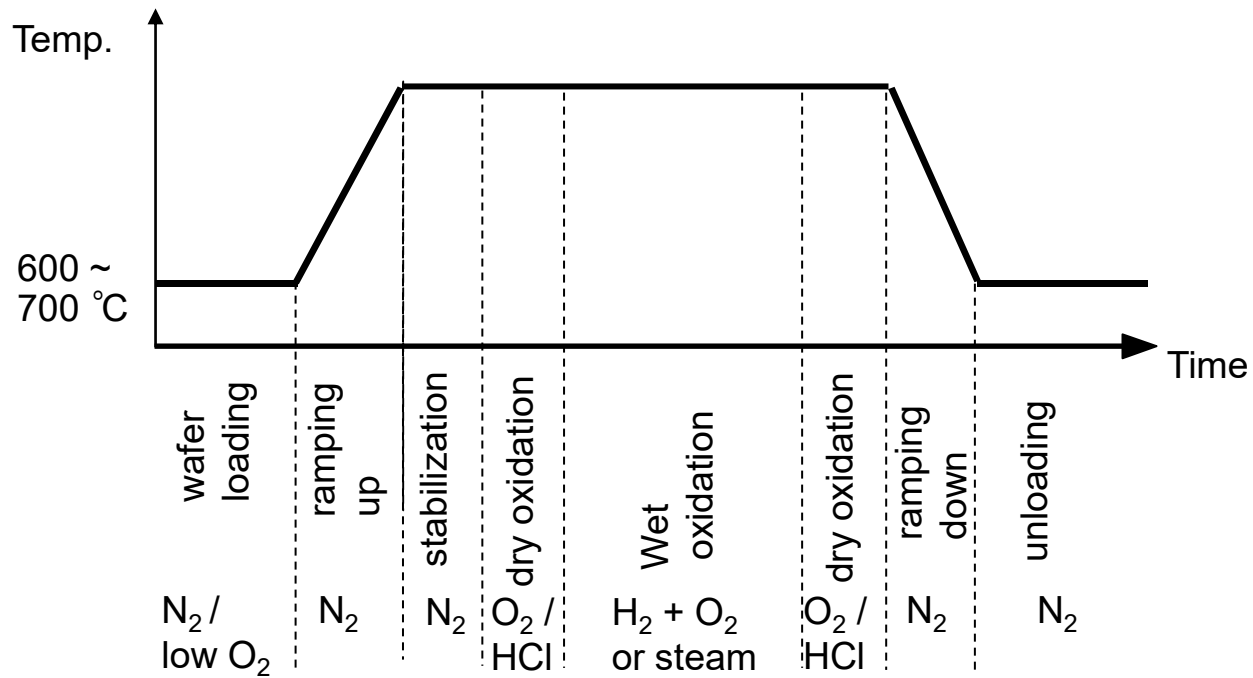
- Recipe chosen depends on the thickness and oxide properties required.
- Thick oxide : usually wet oxidation,
Thin oxide : dry oxidation or very low temperature wet oxidation
- Dry oxidation results in a higher density oxide than that achieved by wet oxide

Dry oxidation

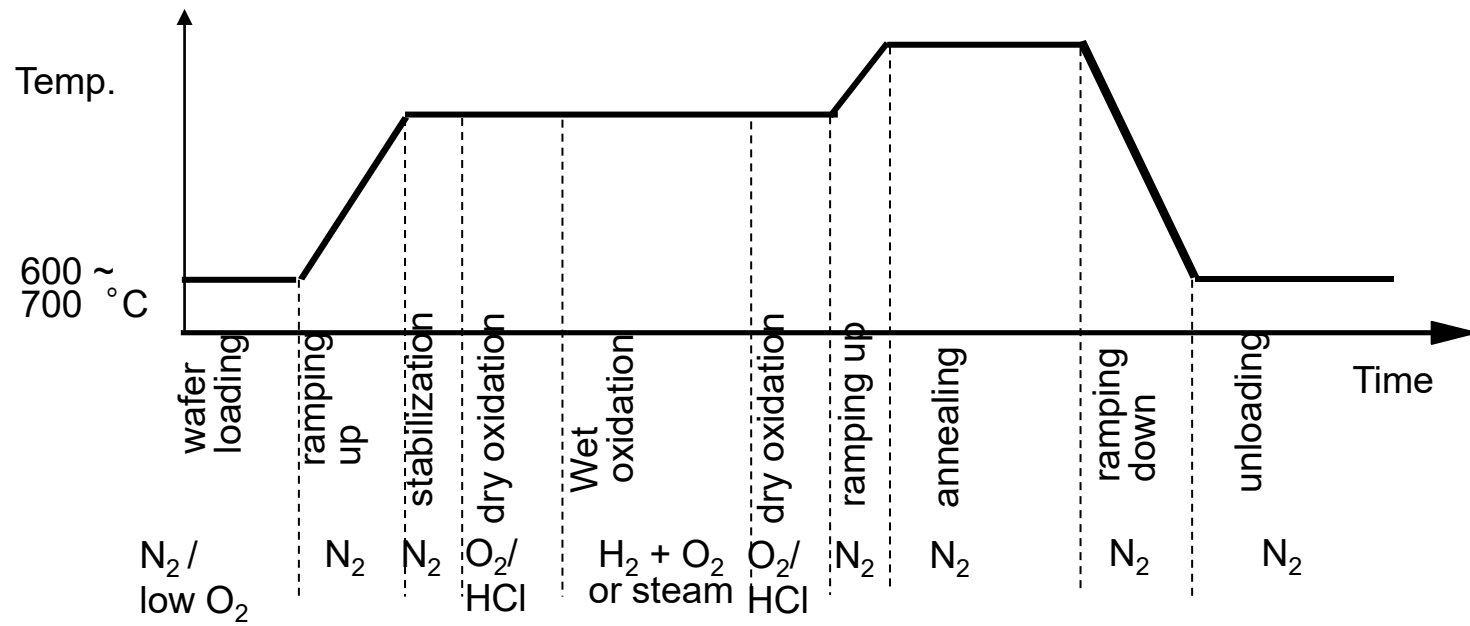


- Ramping up & down rate should be slow enough to ensure no thermal shock to the wafer that can cause warpage or slip dislocation on the wafer.
- Low flow rate O_2 during wafer loading is only for horizontal tube without load lock.

Wet oxidation



Wet oxidation with in-situ annealing

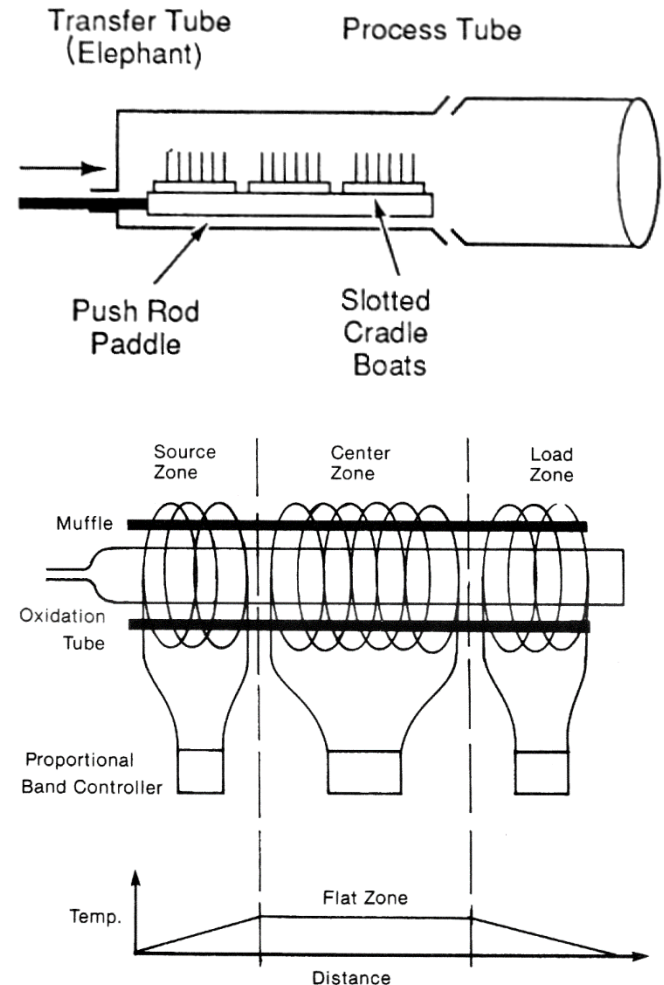


Equipment

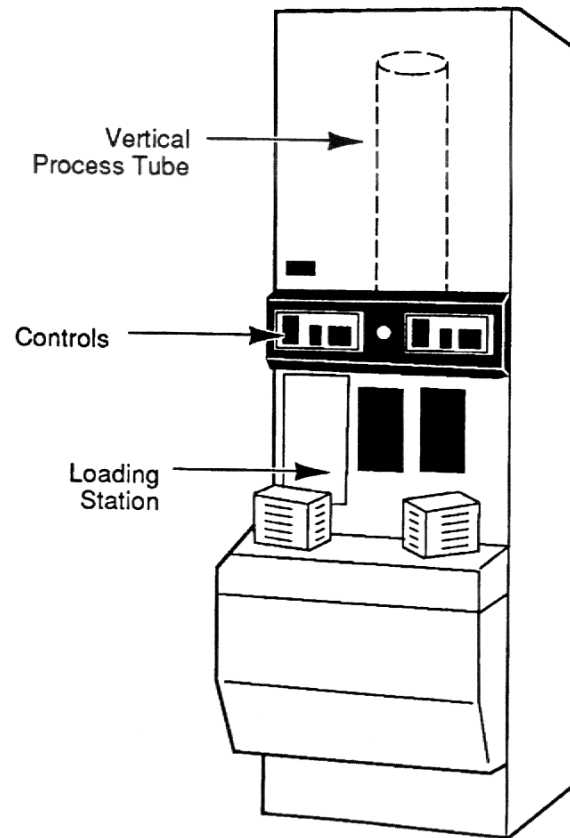
- Horizontal Tube Furnace
- Vertical Tube Furnace

Horizontal Tube Furnace

- Have been used since 1960s for oxidation, diffusion, annealing, and various deposition processes.
- First developed for diffusion process, and to this day are often called simply **diffusion furnaces**.
- The processing wafers are placed on a wafer boat and positioned in the flat zone.
- Temperature deviation in **the flat zone** $\leq \pm 0.5^{\circ}\text{C}$



Vertical Tube Furnace



- Tube is held in a vertical position but wafer is loaded horizontally.
- Widely used for 200 mm and larger wafers